

Attractor Basins of Cooperation, Privacy, and Parasite Persistence:

Applications of *Foundations of Unsupervised Agent Discovery in Raw Dynamical Systems*

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Abstract—Extending Unsupervised Agent Discovery (UAD), we derive analytic conditions under which transparency ($\gamma > 0$) percolates into global cooperation, under which strategic opacity ($\gamma < 0$) preserves privacy, and under which high-entropy “parasites” evade host detection. **Key results:** (i) a configuration-model percolation threshold $\phi_c = \langle d \rangle / (\langle d^2 \rangle - \langle d \rangle)$ bounds the size of the cooperative attractor; (ii) host capacity $C_X^{\text{crit}} = H(A^Y) - \lambda_Y H(I^Y) / \beta$ sets parasite-persistence; (iii) hierarchical links with $\kappa_{iH} < 1$ generate privacy islands despite dense peer cooperation in closed-form expressions.

I. RECAP: BLANKET DISCOVERY AND TRANSPARENCY WEIGHT

Given raw variables $X_i(t)$, Unsupervised Agent Discovery clusters a subset C when

$$I(I_{t+1}^C; E_{t+1}^C \mid S_t^C, A_t^C) \approx 0,$$

yielding sensory S , active A , internal I , external E . Each agent i endows every neighbour j with a signed transparency weight

$$\gamma_{ij} = \frac{\partial}{\partial I(M^j; A_j)} [-\mathbb{E}_q[-\log P(S^i \mid I^i)] - \lambda H(I^i)] \Big|_{I=I_{ij}^*},$$

positive for cooperation, negative for opacity.

II. PAIRWISE COOPERATIVITY INDEX

For ordered pair (i, j) define

$$\kappa_{ij} = \frac{b_{ij} p_{ij} \rho_{ij}}{c_{ij}},$$

with b =benefit per bit shared, p =repeat-interaction probability, ρ =payoff correlation, c =marginal cost. Edge $j \rightarrow i$ is “open” if $\kappa_{ij} > 1$.

$$p(\mathbf{s}_t^i \mid \mathbf{x}_t^i, \mathbf{a}_t^j; \gamma_{ij}) \propto \exp[-\gamma_{ij} d(\mathbf{s}_t^i, \mathbf{a}_t^j)].$$

⁰We build directly on the blanket- and information-theoretic framework of L. et al. (2025), hereafter “UAD.”

III. PERCOLATION THEOREM: GLOBAL ATTRACTOR OF COOPERATION

Let G be a configuration model with degree distribution $P(d)$ and bond-probability $\phi = \Pr(\kappa_{ij} > 1)$. *Theorem.* A giant cooperative component of size $\Theta(N)$ exists iff

$$\phi > \phi_c = \frac{\langle d \rangle}{\langle d^2 \rangle - \langle d \rangle}.$$

If $\phi \leq \phi_c$ all cooperative clusters satisfy $|C| \leq O(\log N)$. Consequently the “attractor basin of cooperation” contains a fraction

$$S(\phi) = 1 - G_0(u(\phi)), \quad u = G_1(1 - \phi + \phi u),$$

with G_0, G_1 the standard generating functions of $P(d)$.

IV. PARASITE-PERSISTENCE CRITERION

A hostile agent Y with action entropy $H(A^Y)$ persists inside host X when

$$C_X < H(A^Y) - \frac{\lambda_Y H(I^Y)}{\beta},$$

where C_X bounds host modelling entropy and β rewards resource extraction. Noise η lowers effective capacity to $C_X - \eta$; adversarial camouflage raises the RHS, enlarging the persistence region.

V. PRIVACY ISLANDS IN HIERARCHIES

For individual i versus hierarchy H we typically have $\kappa_{iH} < 1$. Optimal policy sets $\gamma_{iH} < 0$, driving $I(M^H; S^i) \rightarrow 0$. If a fraction ψ of vertical edges satisfy $\kappa_{iH} < 1$, the network decomposes into:

- 1) a giant peer-cooperative component (if $\phi > \phi_c$),
- 2) disjoint “privacy islands” disconnected from H by closed bonds.

Regulatory or cryptographic measures reduce p_{iH} or raise c_{iH} , expanding ψ and thus the privacy basin.

VI. MIXED AND CONTEXTUAL TRANSPARENCY

Each agent holds a context-dependent $\gamma_{ij}(t)$. Define state $C_{ij}(t)$ tracking reciprocity; update rule

$$\gamma_{ij}(t+1) = \begin{cases} +\gamma_0, & \kappa_{ij}(t) > 1, \\ -\gamma_0, & \kappa_{ij}(t) < 1. \end{cases}$$

Dynamic bond-percolation analysis shows that if the time-average $\bar{\phi}_{ij} = \Pr_t(\kappa_{ij}(t) > 1)$ exceeds ϕ_c the cooperative giant persists in expectation; otherwise cooperation fragments intermittently.

VII. DISCUSSION

The blanket-based disclosure cost rewrites classic Hamilton/Trivers intuitions in measurable bits: *transparency* wins when $\phi > \phi_c$; *privacy* becomes rational when vertical $\kappa < 1$; *parasites* survive whenever host capacity falls below an entropy threshold. These results hold for directed, weighted, or hierarchically nested networks and clarify why higher-complexity organisms (larger C_X) enjoy lower parasite loads yet still value privacy against asymmetric principals.

VIII. CONCLUSION

By marrying UAD with percolation and capacity bounds, we give closed-form conditions for the size of cooperative, private, and parasitic attractors in any large dynamical system—biological, social, or artificial.

IX. COMPARATIVE LITERATURE ON COOPERATION AND PARASITE DYNAMICS

Prior studies on cooperation percolation have addressed binary cooperation-defection edges [1]. Our work significantly extends these by introducing continuous transparency weights derived from mutual information, thus offering a finer-grained analytical perspective. Related empirical findings on parasite-host co-evolution [2] strengthen our theoretical results, providing biological evidence for entropy-based thresholds of parasite persistence. Additionally, the collective intelligence factor [3] supports our percolation thresholds empirically in human groups, confirming transitions in group effectiveness as communication bandwidth varies.

REFERENCES

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